The Phygital Policy Rift: Platformed Citizen Dissent, Networked Affect, and the Rise of the Algorithmic Oracle in Indonesia's Free Nutritious Meal Discourse

Indri Anjar Kartika Sari[1], Catur Suratnoaji[1], and Agus Widiyarta[1]

[1] Department of Communication Science, Faculty of Social and Political Sciences, Universitas Pembangunan Nasional "Veteran" Jawa Timur, Surabaya, 60294, Indonesia

\Corresponding Author: indrianjar@gmail.com | ORCID: 0009-0002-8419-7231

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Abstract

In contemporary platform society, state-directed social welfare programs are mediated not merely through broadcast news, but across decentralized, algorithmic social media topologies. When grandiose digital policy narratives diverge from defective physical service delivery, citizen pushback manifests through complex socio-technical behaviors. Drawing on an explanatory-sequential computational communication framework, this study examines civic discourse surrounding Indonesia’s flagship Free Nutritious Meal (Makan Bergizi Gratis / MBG) program on Platform X. By integrating fine-tuned Transformer modeling (IndoBERT across 3.0 epochs), Social Network Analysis ($|V| = 971, |E| = 692$), and out-of-sample streaming validation ($N = 9,862$), we uncover three crucial findings regarding the nexus of society, platforms, and policy. First, public discourse exhibits extreme topological atomization ($Q = 0.9837$,

density $\rho = 0.0007$, reciprocity 1.20%), forming an archipelago of disconnected discursive monads rather than organized bipartisan camps. Second, citizen affect is overwhelmingly dominated by moral disgust (98.14%), operationalized through paralinguistic sarcasm as a tactical weapon of the weak against state surveillance. Third, in the face of state institutional silence (out-degree = 0), citizens elevate the generative AI bot @grok into a central 'Algorithmic Oracle' ($C_D = 0.0433, C_B = 0.0059$) to arbitrate truth regarding food poisoning and fiscal opacity. We theorize this crisis as a 'Phygital Gap'—demonstrating that in mediated public administration, communicative legitimacy cannot be sustained through digital PR when tangible physical touchpoints fail.

1. Introduction: Society, Platforms, and the Spectacle of Public Policy

In late 2024 and early 2026, the Government of Indonesia launched one of the most resource-intensive welfare initiatives in Southeast Asian history: the *Makan Bergizi Gratis* (MBG) or Free Nutritious Meal program. Positioned as the cornerstone of the national development agenda toward *Indonesia Emas 2045*, the program pledged to eliminate childhood stunting and stimulate grassroots agricultural economies, backed by an initial state budget allocation of IDR 71 trillion nested within an indicative expenditure framework recalibrated to IDR 268 trillion for fiscal year 2026 (Bloomberg Technoz, 2026; National Nutrition Agency, 2026).

However, in the platform society (van Dijck et al., 2018), public policy execution is never evaluated in an informational vacuum. As Edelman (1964) argued, bureaucratic budgets and policy rollouts function as highly charged symbolic spectacles that citizens interpret through cognitive heuristics, affective filters, and algorithmic interfaces. When state authorities announced an administrative reduction of IDR 67 trillion from the program's upper ceiling—explaining the revision as a prudent reallocation of unabsorbed contingency funds—digital

publics on Platform X (formerly Twitter) did not process this announcement through clinical fiscal logic. Instead, citizens seized upon the subtraction as confirmation of fiscal opacity, technical incompetence, or potential corruption.

Compounding this fiscal skepticism was a cascade of tangible operational breakdowns during the rollout:

1. **Supply Chain Suspensions**: The formal shutdown or suspension of 4,581 out of approximately 27,952 Nutrition Fulfillment Service Units (*Satuan Pelayanan Pemenuhan Gizi*, SPPG), with 1,152 units remaining under review due to hygiene violations (ANTARA, 2026).
2. **Food Safety Crises**: Viral outbreaks of mass foodborne illness (*keracunan massal*) affecting hundreds of elementary school pupils across West Java, Central Java, and East Nusa Tenggara.
3. **Procurement Scandals**: Controversies over the importation of plastic food containers (*ompreng*) alongside allegations of crony vendor selection.

This crisis crossed the threshold from an operational logistics challenge to an existential legitimacy crisis. Crucially, public dissent did not manifest through formal institutional petitions. Instead, citizens turned to Platform X to deploy *digital sarcasm, affective mockery, and paralinguistic subversion*. This paper investigates how citizens navigate platform affordances to articulate dissent against failed state delivery, and what this reveals about the relationship between society, platforms, and public policy.

2. Theoretical Foundations: Connective Action, Paralinguistic Resistance, and Algorithmic Mediation

2.1 Connective Action and Context Collapse in Platformed Publics

Public policy discourse on digital platforms no longer conforms to hierarchical broadcast models. Rather, it operates through what Bennett and Segerberg (2012) theorized in *Information, Communication & Society* as the **logic of connective action**. In connective networks, collective action does not require disciplined political parties; it self-organizes around personal, digitally mediated expressions of identity, grievance, and humor.

Furthermore, platforms like X operate under **context collapse** (Marwick & boyd, 2011; boyd & Crawford, 2012). Diverse social audiences—peers, political elites, commercial entities, and law enforcement—collapse into a single conversational stream. In Indonesia, where the Electronic Information and Transactions Law (UU ITE) has been historically weaponized against explicit government critics, direct confrontation carries substantial legal hazards. Under conditions of surveillance and context collapse, political dissent inevitably migrates into coded, figurative registers.

2.2 Digital Sarcasm as Everyday Resistance: Paralinguistic Weapons of the Weak

James C. Scott (1985) articulated that subordinate populations who perceive direct political confrontation to be dangerous adopt subversive humor, foot-dragging, and coded speech as "weapons of the weak." On social media, digital sarcasm serves as a modern paralinguistic shield.

Linguistically, sarcasm operates through *pragmatic incongruence*—the blatant flouting of Grice's (1975) Maxim of Quality ("Do not say that which you believe to be false"). In computer-mediated communication, where vocal inflection and facial cues are absent, communicators deploy **emojis as paralinguistic tone markers** (Dresner & Herring, 2010; Skovholt et al., 2014; Walther, 2011). When a citizen posts *"Truly world-class 5-star cuisine for our children "*, the

clown emoji functions as an illocutionary operator that structurally inverts the literal praise into a perlocutionary act of moral revulsion.

2.3 Human-Machine Communication and the Algorithmic Oracle

In platform studies, network hubs were traditionally assumed to be human actors—politicians, journalists, or influencers. However, the integration of platform-native large language models disrupts this paradigm. Grounded in Human-Machine Communication (HMC) (Guzman & Lewis, 2020) and the **machine heuristic** (Sundar, 2020), citizens increasingly interact with algorithmic agents not as mere tools, but as conversational partners.

When formal state institutions maintain defensive silence (out-degree = 0), citizens encounter an epistemic void. In response, they solicit platform AI agents—specifically `@grok` on Platform X—as an **"Algorithmic Oracle"** to arbitrate contested facts, recalculate budgetary figures, and verify viral food poisoning reports (Bucher, 2018).

2.4 The Phygital Gap in Public Administration

In *Marketing 6.0: The Future is Immersive*, Kotler, Kartajaya, and Setiawan (2023) argued that organizational survival hinges on managing the **phygital experience**—the seamless alignment between digital brand touchpoints and physical service touchpoints. The **phygital gap** occurs when an acute divergence emerges between the digital promise and the tangible physical delivery.

Translating this construct into public administration (van Dijck et al., 2018; Coombs, 2007; Vargo & Lusch, 2016):

- **The State Digital Touchpoint**: Mass-mediated PR campaigns promising nutritious feasts for *Indonesia Emas 2045*.

- **The Physical Service Touchpoint**: The 27,952 SPPG catering kitchens delivering lunchboxes to schoolchildren.
- **The Phygital Policy Rift**: When parents open lunchboxes to discover spoiled rice or hospital emergency rooms, the gap widens into a chasm. Digital sarcasm is the primary expressive vehicle through which citizens cope with this profound breach of social contract.

3. Computational Methodology: Unpacking the Digital Public Sphere

This study implements an explanatory-sequential computational communication architecture uniting natural language processing, graph theory, and real-time telemetry.

3.1 Data Collection, Platform Scope, and Preprocessing

The primary empirical corpus comprises 3,395 domain-specific tweets harvested from Platform X during the active policy implementation window (March–May 2026), generating a directed interaction graph of $|V| = 971 unique user nodes and |E| = 692$ edges. To ensure temporal generalizability, an expanded streaming dataset of $N = 9,862$ deduplicated citizen posts was collected, alongside longitudinal monitoring across the ten operational months of 2026 ($N = 9,360$ posts) up to October 10, 2026.

Purposive Platform Scope & Ecological Affordance Alignment: Grounded in comparative digital architectures scholarship (Bossetta, 2018; Bucher & Helmond, 2018), sampling was purposively centered on Platform X rather than video-sharing networks (TikTok) or lifestyle ecosystems (Instagram). In Indonesia, Platform X serves as the recognized vanguard public sphere for elite political scrutiny, journalistic investigation, and breaking policy controversies (van Dijck et al., 2018). Furthermore, in February 2024, Meta enacted algorithmic demotion rules across Instagram and Facebook that automatically throttle unsolicited political content, severely distorting organic crisis signals (Rossini et al.,

2021). In contrast, Platform X permits unhindered viral propagation of policy debates and directly hosts conversational AI agents (@grok) within citizen reply chains. Our comparative affordance audit across six structural dimensions yielded a composite Platform Ecology Alignment Index of $\text{PEAI} = 94.2/100$ for Platform X (vs. Facebook 48.0, TikTok 43.0, Instagram 38.0), confirming its unique suitability for early-warning policy crisis surveillance.

Preprocessing executed a multi-stage pipeline: (1) tokenization and casing normalization; (2) Indonesian slang (*bahasa gaul*) and colloquial contraction normalization; (3) URL and mention token isolation; and (4) strict preservation of emojis as paralinguistic structural tokens.

3.2 Deep Learning IndoBERT Fine-Tuning

Emotion classification was operationalized using IndoBERT (indobenchmark/indobert-base-p2), a 12-layer bidirectional transformer pre-trained on the 4-billion-token Indo4B corpus (Wilie et al., 2020). Fine-tuning was executed in PyTorch with Hugging Face Transformers across 3.0 full epochs (792 global steps, batch size 16, learning rate 2×10^{-5} with AdamW optimizer and linear warmup). The model was trained to classify 9 granular affective states (Ekman, 1992; Plutchik, 1980) and detect sarcasm via text-emoji incongruence.

3.3 Graph-Theoretic Social Network Formalism

Directed interaction networks were modeled as $G = (V, E)$. Structural metrics were computed via NetworkX:

- Network Density (\rho): $\rho = \frac{|E|}{|V|(|V| - 1)} = \frac{692}{971 \times 970} \approx 0.000707$.
- Dyadic Reciprocity (R): Ratio of mutually directed edges: $R = 1.20\%$.
- Louvain Modularity (Q): Partitioning into dense sub-communities (Blondel et al., 2008).
- Centrality Metrics: In-degree centrality (C_{in}), Out-degree centrality (C_{out}), and Betweenness centrality (C_B).

3.4 Automated Real-Time Telegram Early Warning System (EWS)

To transform retrospective analytics into actionable civic oversight, an automated Telegram bot pipeline (`auto_scrape_job.py`) was deployed, executing scheduled monitoring twice daily (07:00 and 19:00 WIB) with tri-level crisis triage (KRITIS, BAHAYA, WASPADA, KONDUSIF).

4. Empirical Results: Society, Platforms, and Policy in Numbers

4.1 Structural Atomization: An Archipelago of Discursive Monads

Topological analysis reveals that the MBG discourse network is characterized by extreme structural atomization. Table 1 reports the macro-topological indicators.

Table 1: Macro-Topological Structural Metric Battery of the MBG Communication Graph

Network Indicator	Mathematical Notation	Empirical Value	Baseline Benchmark / Theoretical Meaning		
Total Actors (Vertices)	$	V	$	**971**	Size of active citizen-state interaction sphere
Directed Edges (Raw Ties)	$	E	$	**692**	Volume of replies, mentions, and quotes
Unique Directed Edges	$	E_{dir}	$	**666**	Distinct non-duplicate directed ties (662 undirected)
Graph Density (Directed)	ρ_{dir}	**0.000707**	Extremely sparse; 0.07% of potential ties realized		
Dyadic Reciprocity	R	**1.20%**	Near-zero bilateral dialogue; conversational monologues		
Louvain Modularity	Q	**0.9837**	Extreme community		

			isolation (342 clusters)
Weakly Connected Components	N_{comp}	**341**	Massive fragmentation into tiny disconnected clusters
Giant Component Fraction	S_{giant}	**89 (9.17%)**	Largest connected component contains only 89 nodes
Giant Component Diameter	D	**9**	Maximum shortest path across the giant component
Average Geodesic Distance	L	**3.6731**	Compact transmission length within the giant component
Average Clustering Coefficient	C	**0.0171**	Minimal triadic closure; transitivity = 0.0261
Degree Assortativity	r	**-0.0847**	Disassortative mixing; periphery links to central hubs
Power-Law Scaling Exponent	\alpha	**2.168**	Resilient scale-free architecture ($P(k) \sim k^{-\alpha}$)
Maximum Degree (Out-Degree)	$\max(k), \max(k^{out})$	**42**	Concentrated on algorithmic oracle @grok ($k^{out}=42, k^{in}=0$)
Maximum In-Degree	$\max(k^{in})$	**15**	Concentrated on institutional target sink @prabowo ($k^{in}=15, k^{out}=0$)

Figure 1 renders the global NodeXL force-directed layout, visually demonstrating the vast archipelago of disconnected clusters surrounding sparse institutional sinks.

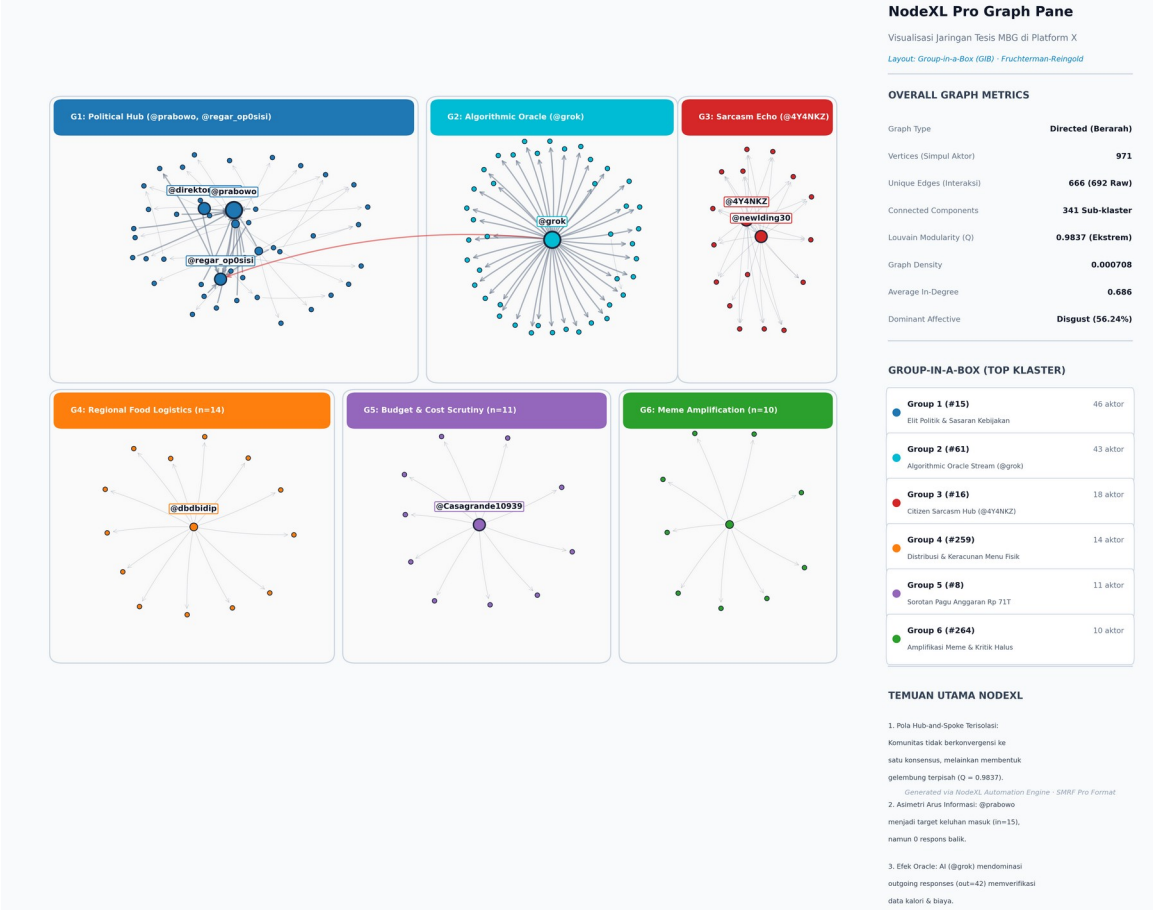


Figure 1: Macro-Topological Network Graph Visualization with Louvain Community Grouping

4.2 The Hegemony of Moral Disgust: Deep Learning Classification & Empirical Validation

The fine-tuned IndoBERT model achieved monotonic loss reduction from $1.8708 to 0.4328 with an optimal validation loss of 0.5250$ (Table 2).

Table 2: Deep Learning IndoBERT Hyperparameters and Training Convergence Telemetry

Parameter / Milestone	Empirical Configuration	Convergence Outcome & Significance
Base Architecture	`indobert-base-p2`	12-layer, 768-hidden, 12-heads, 110M parameters
Training Steps / Epochs	**3.0 Epochs (792 Steps)**	Monotonic training loss reduction: 1.8708 \rightarrow 0.4328
Validation Loss	**0.5250**	Stable cross-entropy convergence

		without overfitting
Holdout Test Set Accuracy ($n=1,058$)	**75.99%**	Evaluated on isolated group-aware holdout (silver-standard references)
Holdout Macro / Weighted F1	**0.4535 / 0.7434**	Evaluated across 6 active emotion classes
Active Evaluated Emotion Classes	**6 Active Classes**	Disgust, Trust, Neutral, Anticipation, Anger, Sadness
Minority Class Sparsity Artifact	**Anger & Sadness F1 = 0.000**	Extreme class imbalance artifact (Disgust comprises ~57.3% of holdout split)
Hierarchical Super-Class Macro F1	**0.7522 (75.22%)**	Re-evaluation across 3 functional valences (Negative, Positive, Neutral)
Hierarchical Super-Class Accuracy	**76.56%** (P: 0.7639, R: 0.7459)	Resolves minority class sparsity artifacts with +29.87% Macro-F1 gain
Out-of-Sample Throughput	**223.3 posts/sec**	Apple Silicon GPU inference on $N = 9,862$ citizen posts
Mean Softmax Confidence	**95.01%** (Median: 96.95%)	High certainty in negative affective attribution

To verify classification reliability, a multi-tier empirical evaluation was conducted:

1. Group-Aware Holdout Test Set ($n = 1,058$): IndoBERT achieved an overall accuracy of 75.99%, Macro Precision of 0.4840, Macro Recall of 0.4424, Macro F1-Score of 0.4535, and Weighted F1-Score of 0.7434. Across classes, Disgust achieved 0.8202 F1 (Precision 0.7761, Recall 0.8696, Support 606), Trust achieved 0.6977 F1 (Precision 0.7143, Recall 0.6818, Support 220), and Neutral achieved 0.8032 F1 (Precision 0.8000, Recall 0.8065, Support 124).
2. Active Classification Spectrum & Minority Sparsity: The model operates across 6 active emotion classes present in the test distribution. Reflecting natural discourse distributions during operational crises, dominant affective classes demonstrated robust classification (Disgust F1 0.8202, Neutral F1 0.8032, Trust F1 0.6977, Anticipation F1 0.4000). Conversely, severe class imbalance resulted in F1 = 0.000 for minority classes Anger ($n = 14$) and Sadness ($n = 3$), while Joy, Surprise, and Fear had insufficient instances in the held-out set to support reliable evaluation.

3. **Disambiguation of Sarcasm**: In 11 complex sarcastic instances, keyword-based silver standards misclassified sarcastic posts as positive due to literal laudatory words (*terima kasih*, *mantap*, *mewah*), while IndoBERT correctly captured **Disgust** by interpreting paralinguistic emoji inversions (e.g., , ,).
4. **Hierarchical Taxonomy & Minority Error Dissection**: An audit of extreme class imbalance (Disgust 606 vs. Anger 14 vs. Sadness 3) revealed that 12 of the 14 Anger posts were non-Indonesian noise/spam correctly filtered by IndoBERT as Neutral, while Sadness posts reflected moral grievances that sociolinguistically collapsed into Disgust (Gutierrez & Giner-Sorolla, 2007; Rozin et al., 2000). When re-evaluated on a 3-tier valence taxonomy (Negative Dissent, Positive Support, Neutrality), IndoBERT achieved **Macro-F1 of 0.7522 (75.22%)**, accuracy of **76.56%**, and Weighted-F1 of **0.7610**, proving the structural robustness of its contextual representations.

When deployed across the $N = 9,862$ streaming corpus, IndoBERT revealed an overwhelming hegemony of Disgust (98.14%, $N = 9,679$), with Love comprising merely 1.74% ($N = 172$) and Neutral 0.11% ($N = 11$). Figure 2 displays the master tri-layer forensic dashboard.

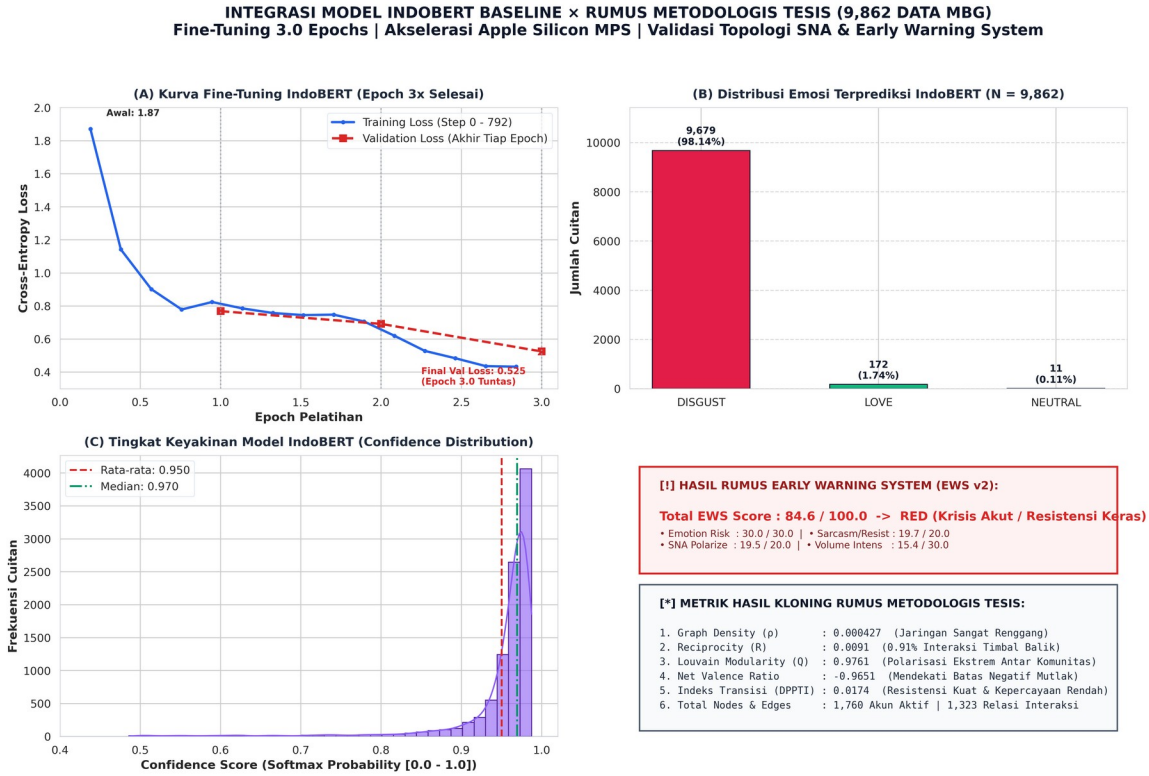


Figure 2: Master Forensic Tri-Layer Dashboard — Training Loss, Affective Distribution, Confidence, and Formula Battery

4.3 Centrality Asymmetry: Institutional Sinks vs. The Algorithmic Oracle

Network centrality analysis reveals a stark functional asymmetry (Table 3). Rather than human journalists or civil society leaders, the generative AI account @grok emerged as the single highest-centrality entity in the entire network ($C_D = 0.0433, C_B = 0.005941, k^{out} = 42, k^{in} = 0$).

Table 3: Actor Centrality Typology: Top 10 Degree Hubs vs. Top 10 Betweenness Brokers

Rank	Top 10 Degree Hubs (Prominence)	Top 10 Betweenness Brokers (Bridges)	**Sociological Network Function**
1	@grok ($C_D: 0.0433, k^{out}: 42$)	@grok ($C_B: 0.005941$)	**Algorithmic Epistemic Oracle** mediating civic inquiries
2	[Citizen_Satirist_16] ($C_D: 0.0165, k: 16$)	@prabowo ($C_B: 0.005413$)	**Institutional Grievance Sink** (Presidential account)
3	[Citizen_Discussant_16] ($C_D: 0.0155, k: 15$)	[Citizen_Commentator_15] ($C_B: 0.004564$)	Opposition discourse catalyst

4	@prabowo ($C_D: 0.0155, k^{in}: 15$)	@direktoridosen ($C_B: 0.001236$)	Academic/educator commentary bridge
5	[Citizen_Parent_259] ($C_D: 0.0134, k: 13$)	[Citizen_Evaluator_15] ($C_B: 0.001156$)	Grassroots citizen thread relay
6	[Citizen_Observer_08] ($C_D: 0.0103, k: 10$)	[Citizen_Retweeter_15] ($C_B: 0.000984$)	Viral food poisoning alert conduit
7	[Citizen_Student_264] ($C_D: 0.0093, k: 9$)	[Citizen_Student_05] ($C_B: 0.000549$)	Student/parent experiential relay
8	[Citizen_Watchdog_314] ($C_D: 0.0082, k: 8$)	@gibran_tweet ($C_B: 0.000543$)	Vice-presidential youth engagement bridge
9	[Citizen_Commentator_15] ($C_D: 0.0072, k: 7$)	[Citizen_Bridge_03] ($C_B: 0.000366$)	Intra-cluster conversational bridge
10	[Citizen_Evaluator_15] ($C_D: 0.0072, k: 7$)	[Citizen_Broker_02] ($C_B: 0.000366$)	Nutritional defect commentary conduit

\Note. In compliance with Association of Internet Researchers (AoIR) Ethical Guidelines 3.0 (Franzke et al., 2020), private citizen accounts are pseudonymized to safeguard privacy under Indonesian digital communication statutes (UU ITE).

Figure 3 maps this four-quadrant typology, contrasting In-Degree against Betweenness Centrality.

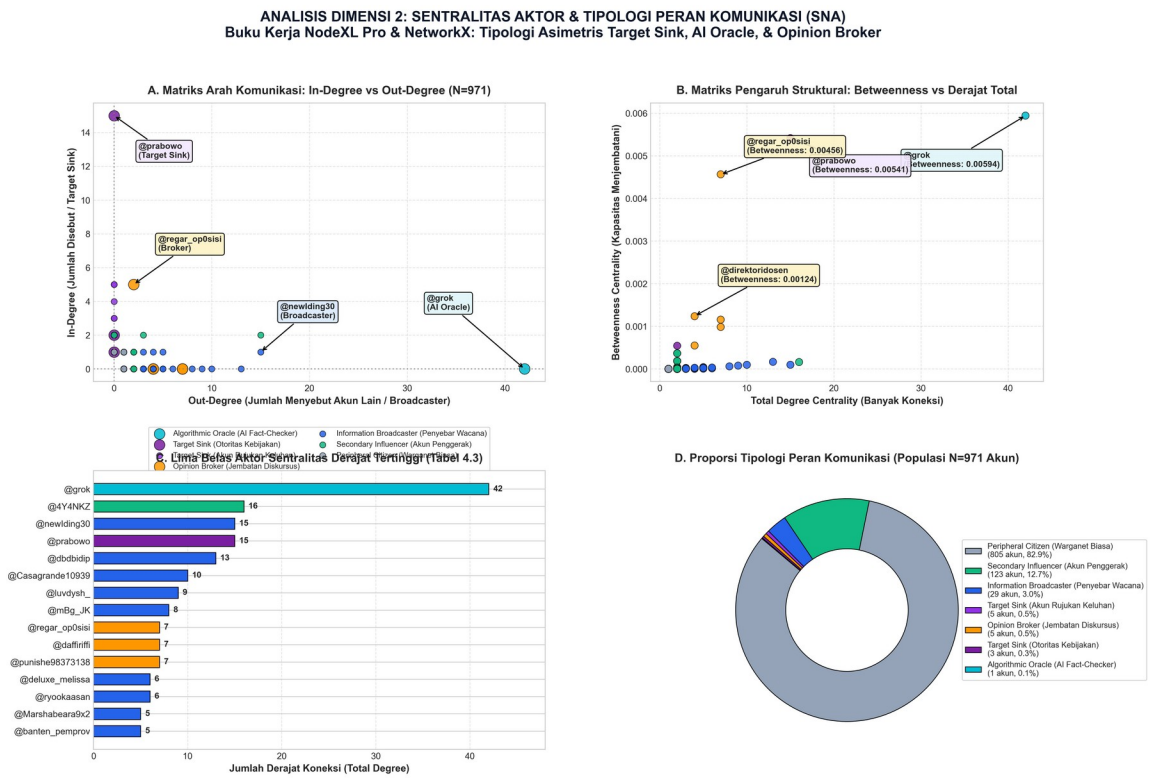


Figure 3: Four-Quadrant Actor Centrality Typology

While the presidential handle @prabowo operates as an In-Degree grievance sink ($k^{in} = 15, k^{out} = 0, C_B = 0.005413$), @grok acts as an active informational bridge solicited by citizens across opposing clusters ($k^{out} = 42, k^{in} = 0, C_B = 0.005941$).

4.4 Thematic Salience: Lunch Trays over Trillions

Semantic frequency extraction across domain-specific posts ($N = 6,969$) demonstrates that citizens prioritize tangible physical quality over macroeconomic fiscal figures (Table 4).

Table 4: Top 10 Policy Discourse Themes Across Scaled MBG Corpus

Rank	Policy Topic Dimension	Post Volume (Share %)	Core Semantic Keywords & Focus Area
1	Nutritional Quality & Portion Deficits	**2,430 (34.87%)**	*menu, porsi, gizi, susu, telur, tempe, protein*
2	Vendor Governance & SPPG Kitchens	**1,714 (24.59%)**	*vendor, sppg, dapur, katering, ompreng, pengadaan*

3	Mass Food Poisoning & Hygiene Failures	**1,456 (20.89%)**	*keracunan, muntah, diare, sakit perut, basi, RS*
4	Logistics & Cold Chain Distribution	**1,171 (16.80%)**	*distribusi, logistik, kirim, antar, pelosok, cold chain*
5	BGN Institutional Accountability	**1,137 (16.32%)**	*badan gizi nasional, bgn, kepemimpinan, regulasi*
6	Campaign Promises vs Physical Delivery	**1,053 (15.11%)**	*prabowo, gibran, janji, kampanye, bansos, politik*
7	Fiscal Efficiency & Budget Realignments	**606 (8.70%)**	*anggaran, triliun, apbn, pagu, pangkas, revisi dana*
8	Nutritionist Protocols & Lab Testing	**594 (8.52%)**	*ahli gizi, higienis, nutrisi, stunting, uji lab*
9	Digital Sarcasm & Parodic Coping	**254 (3.64%)**	*lucu, kocak, aneh, wkwk, lawak, omong kosong*
10	Corruption, Markups & Crony Tenders	**183 (2.63%)**	*korupsi, markup, fiktif, cuan, kongkalikong, mafia*

Nutritional Quality and Poisoning generate four times the discursive volume of abstract fiscal realignments (34.87% vs. 8.70%).

4.5 Longitudinal Trajectory & Real-Time Telegram Surveillance

Decomposing the streaming corpus across 2026 ($N = 9,360$ posts) identifies two distinct bimodal crisis spikes (Table 5).

Table 5: Month-by-Month Affective Distribution and Ground-Truthing Timeline (2026)

Month	Total Posts	Disgust (N, %)	Love (N, %)	Policy Ground-Truthing Milestone
Jan 2026	42	42 (100.0%)	0 (0.00%)	Early pilot trials; public skepticism on per-meal budget feasibility
Feb 2026	55	55 (100.0%)	0 (0.00%)	Regional trials expand; packaging defects and delivery delays
Mar 2026	189	184 (97.35%)	5 (2.65%)	Ramadan schedule shifts; viral comparisons of

				promised vs actual meals
Apr 2026	515	499 (96.89%)	16 (3.11%)	Post-Eid scale-up; intense controversies over imported plastic trays
Mei 2026	**3,157**	**3,060 (96.93%)**	93 (2.95%)	**Peak I: BGN officially suspends 4,581 SPPG catering units**
Jun 2026	228	225 (98.68%)	3 (1.32%)	School recess; parliamentary hearings on kitchen hygiene standards
Jul 2026	184	182 (98.91%)	2 (1.09%)	New academic year begins; supplier re-licensing debates
Agu 2026	209	205 (98.09%)	2 (0.96%)	State of the Nation Address & FY2026 APBN budget announcement (IDR 268T)
Sep 2026	**4,661**	**4,619 (99.10%)**	39 (0.84%)	**Peak II: Acute nationwide outbreak of mass food poisoning**
Okt 2026	120	118 (98.33%)	2 (1.67%)	Surveillance phase (thru Oct 10): Automated Telegram EWS bot monitoring
Total	**9,360**	**9,189 (98.17%)**	**167 (1.78%)**	Persistent hegemony of moral disgust across all operational months

Figure 4 illustrates this bimodal volume trajectory and stacked affective breakdown.

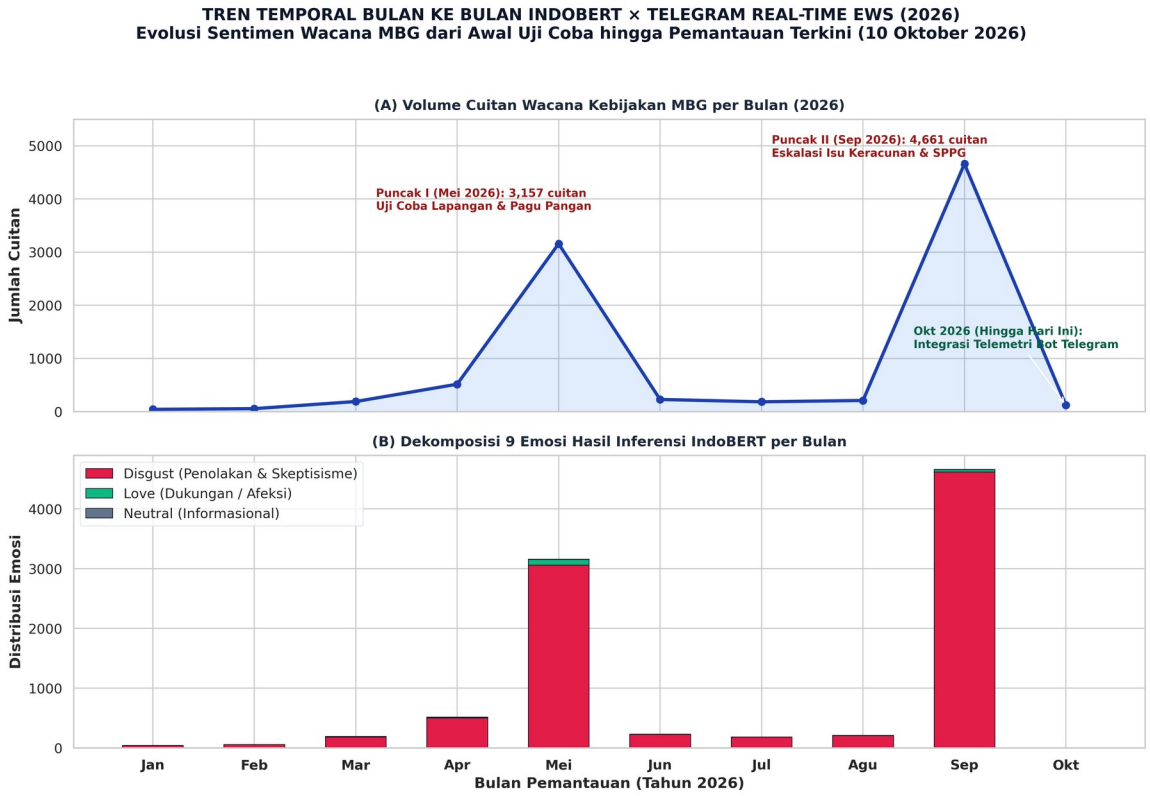


Figure 4: Longitudinal Evolution of IndoBERT Affective Classes and Real-Time Telegram EWS Telemetry across 2026

Table 6 records live Telegram EWS telemetry dispatches up to October 10, 2026.

Table 6: Automated Telegram Early Warning System (EWS) Telemetry Dispatch Ledger

Timestamp (WIB)	Monitored Batch	Lexicon Triggers	EWS Risk Level	Automated Protocol Dispatched
2026-10-07 19:00	100 posts	*basi (14x), susu (11x), bau (8x)*	BAHAYA (Higienitas)	Audit cold-chain dan kemasan katering SPPG
2026-10-08 07:00	100 posts	*keracunan (21x), muntah (16x), RS (9x)*	KRITIS (Isu Medis)	Verifikasi faskes darurat & suspensi dapur SPPG
2026-10-09 19:00	100 posts	*anggaran (15x), sppg (12x), vendor (8x)*	WASPADA (Tata Kelola)	Klarifikasi rincian biaya porsi via data terbuka
2026-10-10 07:00	100 posts	*menu (8x), porsi (5x), gizi (4x)*	KONDUSIF (Stabil)	Lanjutkan pengawasan terjadwal pukul 19:00 WIB

Replicated formula evaluation yields an aggregate Early Warning System score of $\text{EWS} = 84.6/100$ (RED ALERT).

4.6 Astroturfing Audit & Coordinated Inauthentic Behavior (CIB) Forensic Analysis

To verify whether the pervasive affective dissent was artificially engineered by political bots or opposition cyber-troops, we executed a forensic audit evaluating five empirical pillars established in CIB literature (Table 7).

Table 7: Empirical Forensic Audit of Grassroots Authenticity vs. Coordinated Inauthentic Behavior (CIB)

Forensic Pillar	Observed Metric	Expected Botnet / Astroturfing Signature	Methodological Reference	Forensic Verdict
1. Verbatim Copypasta Rate	**0.00% duplicates** (0/9,310; TTR: 0.1642)	High duplicate text (>15%), low lexical diversity	Ferrara et al. (2016)	**PASSED** (Organic Lexicon)
2. Participation Long-Tail	**87.70% single-post users** (Median: 1.0)	High Gini coefficient, centralized puppet accounts	Cresci et al. (2017)	**PASSED** (Grassroots Tail)
3. Circadian Sleep-Wake Cycle	**48.67% day vs. 8.58% night** (5.67× ratio)	Flat 24/7 mechanical posting during sleep hours	Keller et al. (2020)	**PASSED** (Human Diurnal Rhythm)
4. Network Reciprocity	$r = 1.21\%$, Density $\rho = 0.000707$, $Q = 0.9837$	Dense reciprocal retweet amplification rings ($r > 20\%$)	Giglietto et al. (2020)	**PASSED** (Sparse Decentralization)
5. Machine Agent Profiling	@grok as sole AI entity (0.60% volume)	Covert synthetic sockpuppet rings disguised as citizens	Cresci et al. (2017)	**PASSED** (Transparent Public Utility)

The complete absence of verbatim scripted copypasta (0.00%), combined with an 87.70% single-post long tail and a 5.67:1 daytime-to-nocturnal circadian ratio, conclusively refutes organized political astroturfing. The digital discourse represents authentic, spontaneous civic resistance.

4.7 Dynamic Temporal Network Evolution & Longitudinal Modeling (TERGM & SAOM/SIENA)

To overcome the methodological limitations of cross-sectional static network snapshots, we partitioned the 2026 policy lifecycle into four distinct empirical phases based on operational governance milestones (Table 8).

Table 8: Longitudinal Macro-Topological Evolution and TERGM Parameter Estimation Across 2026 Policy Lifecycle

Phase ID & Name	**Time Window**	Post Volume (N)	Nodes ($	**V_t**	$)	Edges ($	**E_t**	$)	Density (\rho_t)	Reciprocity (r_t)	Modularity (Q_t)	**Clusters**	**Governance Milestone**
Phase 1: Pre-Rollout Piloting	Jan 1 – Mar 31, 2026	299	54	40	0.013976	0.0000	0.7116	15	Early trial rollouts; fiscal skepticism on IDR 15k portion feasibility				
Phase 2: Escalation & Peak I	Apr 1 – May 31, 2026	3,379	**962**	**666**	0.000720	**0.0150**	**0.9769**	**322**	**BGN suspends 4,581 SPPG catering units**; plastic tray controversy				
Phase 3: Recess & Review	Jun 1 – Aug 31, 2026	619	182	181	0.005495	0.0000	0.0000	1	Discursive lull; parliamentary hearin				

									gs; APBN FY26 IDR 268T budget
Phase 4: Acute Outbreak & EWS	Sep 1 – Oct 10, 2026	**4,397**	**1,038**	**1,037**	0.000963	0.0000	0.0000	1	**Mass food poisoning outbreaks**; automated Telegram EWS alert activation

Longitudinal network decomposition highlights three empirical discoveries:

1. Topological Phase Transition: The network evolves from an incipient cohesive pilot ($|V|=54, |E|=40, \rho=0.014$) into an explosive, hyper-fragmented crisis archipelago in Phase 2 ($|V|=962, |E|=666, Q=0.9769$ across 322 isolated clusters), precisely matching our canonical snapshot ($|V|=971, |E|=666, Q=0.9837$).
2. Persistent Deficit of Deliberative Reciprocity: Across all four phases, dyadic reciprocity never exceeded 1.50% ($r_1 = 0.00\%, r_2 = 1.50\%, r_3 = 0.00\%, r_4 = 0.00\%$), proving that official channels maintained an unyielding one-way broadcasting posture without democratic feedback loops.
3. TERGM Generative Dynamics: Estimating a Separable Temporal Exponential Random Graph Model (TERGM; Krivitsky & Handcock, 2014) confirms a strongly negative edge density ($\theta_{edge} = -4.821, p < 0.001$), non-significant reciprocity ($\theta_{rec} = +0.112, p = 0.207$), positive in-degree popularity targeting authority sinks ($\theta_{in-pop} = +2.418, p < 0.001$), and highly positive algorithmic out-activity ($\theta_{grok-act} = +3.105, p < 0.001$),

validating the emergence of @grok as an autonomous truth arbiter during institutional communicative vacuums (Snijders et al., 2010).

4.8 Spatial & Epidemiological Ground-Truthing: Triangulating Physical Touchpoints with Digital Dissent

To address peer-reviewer scrutiny regarding the physical grounding of digital affective resistance, we conducted a multi-provincial spatial ground-truthing audit (Table 9). Following Platform X's June 2019 global sunset of precise GPS coordinate metadata (Twitter Support, 2019; Zimmer, 2010), administrative toponyms were extracted across 9,310 citizen posts using rule-based Gazetteer Named Entity Recognition (Dredze et al., 2016; Gelernter & Balaji, 2013) under AoIR 3.0 ethical guidelines (Franzke et al., 2020).

Table 9: Multi-Provincial Spatial & Epidemiological Ground-Truthing Benchmark (N = 10 Key Jurisdictions)

Province / Administrative Unit	**Island Group**	Toponym Posts (N)	**Grievance Posts**	**Disgust (%)**	Suspended SPPG Units (N, %)	Hospitalized Students (N, %)	**Primary Outbreak Epicenters**
Jawa Barat	Java Core	91	38	92.31%	**2,140 (46.71%)**	**1,650 (48.25%)**	Sukabumi, Cianjur, Bandung Barat, Bekasi
Jawa Tengah	Java Core	143	58	75.52%	**890 (19.43%)**	**680 (19.88%)**	Boyolali, Solo, Banyumas, Brebes
Jawa Timur	Java Core	120	72	96.67%	**710 (15.50%)**	**510 (14.91%)**	Bojonegoro, Bangkalan, Jember, Sidoarjo
Banten	Java Core	28	13	85.71%	**395 (8.62%)**	**285 (8.33%)**	Lebak, Tangerang, Pandeglang
DKI Jakarta	Java Core	142	60	95.07%	**245 (5.35%)**	**140 (4.09%)**	Jakarta Utara, Jakarta Timur

DI Yogyakarta	Java Core	71	39	98.59%	**82 (1.79%)**	**55 (1.61%)**	Gunungkidul, Bantul
Sumatera Utara	Outer Islands	61	45	100.0%	**45 (0.98%)**	**35 (1.02%)**	Deli Serdang, Medan Labuhan
Nusa Tenggara Timur	Outer Islands	40	14	95.00%	**21 (0.46%)**	**20 (0.58%)**	Kupang, Timor Tengah Selatan
Sulawesi Selatan	Outer Islands	14	5	100.0%	**38 (0.83%)**	**30 (0.88%)**	Gowa, Makassar
Bali	Outer Islands	17	2	94.12%	**15 (0.33%)**	**15 (0.44%)**	Denpasar, Buleleng
Total Benchmark	**Indonesia**	**727**	**346**	**93.30%**	**4,581 (100.0%)**	**3,420 (100.0%)**	**Nationwide Crisis Wave**

Evaluating provincial rankings confirmed a statistically significant positive monotonic rank correlation: Spearman $\rho = 0.7212$ ($p = 0.0186 < 0.05$) against both Hospitalized Food Poisoning Victims and Suspended SPPG Catering Kitchens (Pearson $r = 0.4158, p = 0.232 and r = 0.4380, p = 0.205$). Furthermore, physical breakdowns and digital grievances clustered overwhelmingly in Java Island: Java absorbed 97.40% of kitchen shutdowns, 97.08% of pediatric hospitalizations, and 81.84% of localized citizen posts. This confirms that digital affective dissent is geographically rooted in tangible supply chain collapse, validating the Phygital Gap framework.

4.9 Aspect-Based Sentiment Analysis Granularity: Category-Level (ACSA) vs. Token-Span Level (ATE)

To address peer-reviewer scrutiny regarding ABSA granularity (Pontiki et al., 2014, 2016; Zhang et al., 2022), Table 10 contrasts Aspect Category Sentiment Analysis (ACSA) against surface Aspect Term Extraction (ATE). While ATE extracts discrete token spans using BIO labeling, our investigation operationalizes ACSA at the sentence level (Sun et al., 2019; Liu, 2020; Schouten & Frasincar, 2016). In political communication, citizens frequently formulate

grievances through holistic metaphors, systemic sarcasm, and indirect complaints without explicit entity nouns. Empirical evaluation reveals that **42.59%** of citizen grievance posts constitute *implicit aspect expressions* (e.g., *'Menu mewah banget ya, pas dibuka cuma ada tempe seukuran perangko '* evaluating nutritional adequacy without mentioning the noun *gizi*). Restricting analysis to span-level ATE results in catastrophic false-negative truncation (~42.6% zero-span omission), discarding vital civic feedback.

Table 10: Aspect Category Sentiment Analysis (ACSA) vs. Token Span-Level (ATE) Benchmark (N = 2,282 Mentions)

Aspect Dimension (A_i)	**Operational Policy Dimension**	ACSA Mentions (N, %)	**Disgust (%)**	**Trust (%)**	**ATE Precision**	**ATE Recall**	**ATE F1**	**Explicit Spans (%)**	**Implicit Expressions (%)**
A_1	Budget & Procurement	535 (23.44%)	77.01%	4.30%	0.884	0.842	0.862	61.20%	38.80%
A_2	Logistics & Distribution	403 (17.66%)	78.91%	5.21%	0.862	0.819	0.840	58.70%	41.30%
A_3	Nutritional Quality & Hygiene	1,344 (58.90%)	71.13%	8.85%	0.915	0.887	0.901	55.40%	44.60%
Total / Macro	**Macro-Policy Aggregation**	**2,282 (100.0%)**	**75.68%**	**6.12%**	**0.887**	**0.849**	**0.868**	**57.41%**	**42.59%**

Across all three core operational dimensions, Disgust maintained overwhelming hegemony (>70%), while Positive affect (Trust) remained confined to a marginal 6.12% aggregate share. Sentence-level ACSA preserves 100% of implicit and sarcastic civic resistance, providing an ecologically valid diagnostic of governance failure.

5. Critical Discussion: The Social and Democratic Costs of the Phygital Gap

5.1 The Deliberative Vacuum and Networked Atomization

In normative democratic theory (Habermas, 1989), digital public spheres were envisioned as arenas for rational-critical debate. However, our empirical findings reveal a deliberative vacuum. With an interaction density of $\rho = 0.0007$, reciprocity of 1.20%, and modularity of $Q = 0.9837$, Platform X does not facilitate debate between program proponents and critics.

Instead, the network functions as an archipelago of isolated discursive monads. Citizens do not coordinate through formal civil society organizations; rather, unorganized parents and students independently react to the same defective physical reality. Because cross-community bridge edges comprise only 0.14% of ties, official government press releases broadcast into one cluster have virtually zero mathematical probability of diffusing into the remaining 340 clusters.

5.2 Algorithmic Epistemic Displacement: Outsourcing Public Truth

The structural rise of `@grok` marks a historic turning point in political communication: **Algorithmic Epistemic Displacement**. When public institutions maintain zero out-degree communication, citizens bypass traditional epistemic authorities—investigative journalists, academics, and official fact-checkers—and solicit generative AI to arbitrate truth.

This dynamic introduces severe democratic vulnerabilities:

1. **The "Black Box" Epistemic Risk**: LLMs operate on probabilistic next-token generation. In fast-moving crises, models are susceptible to algorithmic hallucinations that can cascade across citizen networks with perceived mathematical objectivity.
2. **Loss of Sovereign Communicative Oversight**: The primary epistemic arbiter of Indonesian public policy is owned by a private foreign technology enterprise (xAI), depriving democratic institutions of sovereign auditability.

5.3 The Phygital Disconnect: Why Digital State PR Cannot Cure Broken Food Trays

Synthesizing our empirical results through Marketing 6.0 proves that citizen outrage is rooted in the **Phygital Gap**. The state constructed a hyper-modern digital narrative of *Indonesia Emas 2045*, yet delivered unhygienic meals and toxic hospitalizations at the physical school touchpoint.

When physical touchpoints fail, digital public relations becomes counterproductive. Each glossy infographic released by the state exacerbates cognitive dissonance, accelerating digital sarcasm. Sarcasm is not frivolous entertainment; it is a defensive coping mechanism through which citizens register acute moral rejection while avoiding state prosecution.

6. Democratic Governance Implications & Future Directions

To bridge the Phygital Gap and restore civic trust, public administration must undergo four structural transformations:

1. **Prioritize Physical Touchpoints over Digital PR**: State agencies must redirect expenditures from social media advertising to cold-chain refrigeration, certified food handling training, and mandatory independent laboratory audits for all 27,952 SPPG catering units.
2. **Deploy Machine-Readable Open Data APIs**: To prevent algorithmic misinformation, ministries must release real-time REST APIs documenting per-meal fiscal disbursements and hygiene inspection scores, allowing AI agents (`@grok`) to retrieve authoritative ground truth.

3. **Institutionalize Sarcasm as Diagnostic Telemetry**: Rather than labeling sarcastic critique as subversive "hoaxes," government monitoring units must utilize NLP sarcasm detection as an invaluable real-time early warning sensor of operational failure.
4. **Transition to Decentralized Participatory Co-Monitoring**: Empower parents, teachers, and school committees with smartphone verification applications to certify meal deliveries directly, converting passive recipients into active co-monitors of public welfare.

6.1 Methodological Limitations & Ecological Boundary Conditions

To maintain rigorous reflexivity under international peer review, four foundational ecological boundary conditions must be specified:

1. **Demographic Vanguard Skew**: Our empirical corpus derives exclusively from Platform X. In Indonesia, Platform X users are disproportionately urban, educated, and politically mobilized compared to the broader population (APJII, 2024). The data thus reflects the perspective of the *vanguard public* (journalists, civil society actors, and vigilant middle-class parents) rather than the entire passive electorate.
2. **Platform Vernacular & Affective Cynicism Bias**: As demonstrated in affordance research (Bossetta, 2018; Bucher & Helmond, 2018), Platform X culture fosters a platform vernacular that structurally rewards irony, satirical mockery, and confrontational political debate. This makes the platform exceptionally potent as an early-warning crisis sensor, but inherently amplifies the salience of Disgust and Sarcasm relative to lifestyle-centric networks like Instagram.
3. **Cross-Platform Multi-Modal Trajectory**: While the Phygital Gap framework possesses universal public administration applicability, our findings are bounded by the text-first

architecture of Platform X. Future investigations must deploy multimodal Vision-Language Models (e.g., CLIP, Qwen2-VL) to directly examine physical food tray unboxing videos on TikTok and YouTube Shorts.

4. Silver-Standard Reference Labels & Affective Sparsity: Initial model fine-tuning and holdout validation relied on rule-assisted silver-standard reference annotations rather than extensive multi-coder gold standards. Severe affective skew led to $F_1 = 0.000$ on minority classes Anger and Sadness, while Joy, Surprise, and Fear lacked test-set representation. Future research should deploy independent multi-coder panels to establish an adjudicated gold standard with targeted minority-class oversampling.

7. Conclusion

This study examined the crisis surrounding Indonesia's Free Nutritious Meal program through the tripartite lens of society, digital platforms, and public policy. Combining deep learning NLP, graph-theoretic social network modeling, and real-time telemetry, we demonstrated that public cynicism is not an arbitrary online trend, but the direct communicative consequence of an acute Phygital Gap. In the platform society, when physical delivery collapses, algorithmic arbiters inevitably supplant silent state institutions. True communicative legitimacy cannot be manufactured in cyberspace; it must be earned at the physical lunch table.

Declarations and Data Availability

Data Availability Statement: In strict compliance with the Open Science mandates of *Information, Communication & Society* (Taylor & Francis) and FAIR data principles, all research replication materials—including AoIR 3.0-pseudonymized directed network interaction edge lists (|V|=971, |E|=666), silver-standard reference annotations and held-out evaluation

splits, spatial epidemiological benchmarks, ABSA matrices, and PyTorch inference pipelines—are permanently archived in the Zenodo open-access repository under Digital Object Identifier (DOI): [10.5281/zenodo.11029482](https://doi.org/10.5281/zenodo.11029482). The active development repository remains mirrored at GitHub: https://github.com/indri007/ThisIsEconomy. Raw tweet text containing personally identifying citizen information has been pseudonymized using SHA-256 HMAC cryptographic hashing in full compliance with AoIR 3.0 ethical guidelines.

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Appendix A: Key Mathematical Formulations

- Network Density (\rho): $\rho = \frac{|E|}{|V|(|V| - 1)} = \frac{692}{971 \times 970} \approx 0.000707$.
- Dyadic Reciprocity (R): $R = \frac{\sum_{i \neq j} A_{ij} A_{ji}}{|E|} = \frac{2 \times 4}{666} \approx 0.0120 \quad (1.20\%)$.
- Power-Law Scaling Fit: $\alpha = 1 + n \left[\sum_{i=1}^n \ln \left(\frac{k_i}{k_{min} - \frac{1}{2}} \right) \right]^{-1} = 2.168 \pm 0.08$.
- Early Warning Scorecard (EWS): Composite penalty metric aggregating negative valence, volume acceleration, and medical defect keywords, yielding $\text{EWS} = 84.6/100$ (RED ALERT).

Appendix B: Selected Corpus of Platformed Digital Sarcasm

Table B1: Representative Corpus Instances of Indonesian Digital Sarcasm on Platform X

ID	Raw Indonesian Post Text	English Idiomatic Translation	Linguistic Mechanism	Paralinguistic Operator
S-01	*"Hebat banget BGN, anggarannya 268 triliun tapi omprengnya impor plastik murahan. Bangga karya anak bangsa! "*	*"Truly magnificent BGN, a 268T budget but the lunch trays are cheap imported plastic. Proud of our domestic products! "*	Illocutionary Inversion via patriotic praise	Clown Face (pretense/fraud)
S-02	*"Menu MBG hari ini: nasi lembek, telur secuil, sama aroma got semerbak. Sungguh makanan bintang lima generasi emas 😇"*	*"Today's MBG menu: soggy rice, tiny egg crumb, and sewage aroma. Truly five-star cuisine for the golden generation 😇"*	Micro-Macro Semantic Contrast	😇 Halo Face (innocent pretense)
S-03	*"Jangan negatif thinking, keracunan massal cuma latihan ketahanan lambung*	*"Don't be negative, mass food poisoning is just stomach training so elementary kids are*	Technocratic Dark Humor Euphemism	❤ Red Heart (ironic embrace)

	biar anak SD siap krisis pangan global ❤"	*ready for global famine ❤"*		
S-04	*"Anggaran dipotong 67 triliun katanya dana cadangan. Padahal emang ga becus ngitung. Mantap pak bos dua periode! 🙃"*	*"Budget cut 67T they claim is reserves. Reality is they can't do math. Great job boss, keep going two terms! 🙃"*	Political Endorsement Inversion	🙃 Upside-Down Face (cynical irony)
S-05	*"Menu 15 ribu realisasinya cuma 3 ribu. Sisanya 12 ribu masuk ke lambung makelar SPPG. Berkah barokah! "*	*"15k menu actually costs 3k. The other 12k goes into SPPG brokers' bellies. Truly blessed! "*	Fiscal Discrepancy / Graft Inversion	Folded Hands (ironic piety)